Vectors and Embeddings in AI

ANSWER KEY

Section 1: Vector Spaces and Linear Combinations

1 Given word vectors $\mathbf{u} = [2, -1, 3]^T$ and $\mathbf{v} = [1, 4, -2]^T$, compute the linear combination $3\mathbf{u} - 2\mathbf{v}$.

a. $[4, -11, 13]^T$ ✓

b. $[4, 5, 5]^T$

c. $[8, -11, 5]^T$

d. $[7, 5, 13]^T$

$3[2, -1, 3] = [6, -3, 9]$. $2[1, 4, -2] = [2, 8, -4]$. Difference: $[6-2, -3-8, 9-(-4)] = [4, -11, 13]^T$.

2 Two embedding vectors $\mathbf{a} = [k, 2, -1]^T$ and $\mathbf{b} = [3, -4, 2]^T$ are orthogonal in semantic space. Find the value of k.

a. $k = \frac{10}{3}$ ✓

b. $k = 2$

c. $k = -\frac{10}{3}$

d. $k = 5$

Orthogonal $\Rightarrow \mathbf{a} \cdot \mathbf{b} = 3k - 8 - 2 = 3k - 10 = 0 \Rightarrow k = 10/3$.

3 Determine whether the three feature vectors $\mathbf{v}_1 = [1, 0]^T$, $\mathbf{v}_2 = [0, 1]^T$, and $\mathbf{v}_3 = [3, -2]^T$ are linearly independent in $\mathbf{R}^2$.

a. **Linearly dependent (any set of > 2 vectors in $\mathbf{R}^2$ is dependent)** ✓

b. Linearly independent

c. Orthogonal set

d. Cannot be determined without a matrix inversion

Since the dimension of the space is 2, any collection of 3 vectors is linearly dependent: $\mathbf{v}_3 = 3\mathbf{v}_1 - 2\mathbf{v}_2$.

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Section 2: Vector Norms and Sparsity

4 Calculate the Manhattan (L_1) norm and Euclidean (L_2) norm of the embedding error vector $\mathbf{e} = [3, -4, 12]^T$.

a. $\|\mathbf{e}\|_1 = 19, \quad \|\mathbf{e}\|_2 = 13$ ✓

b. $\|\mathbf{e}\|_1 = 13, \quad \|\mathbf{e}\|_2 = 19$

c. $\|\mathbf{e}\|_1 = 11, \quad \|\mathbf{e}\|_2 = 13$

d. $\|\mathbf{e}\|_1 = 19, \quad \|\mathbf{e}\|_2 = 169$

$L_1 = |3| + |-4| + |12| = 19$. $L_2 = \sqrt{3^2 + (-4)^2 + 12^2} = \sqrt{9 + 16 + 144} = \sqrt{169} = 13$.

5 Find the Chebyshev (L_∞) norm of the parameter vector $\mathbf{w} = [-7, 2, 5, -9, 4]^T$.

a. 9 ✓

b. 7

c. 27

d. 5

$\|\mathbf{w}\|_\infty = \max(|-7|, |2|, |5|, |-9|, |4|) = 9$.

6 In Lasso regression, why does adding an L_1 penalty to the loss function induce exact feature sparsity (weights driven to 0)?

a. **The diamond-shaped L_1 contour has sharp corners on the coordinate axes** ✓

b. The L_1 norm is continuously differentiable everywhere

c. The L_1 penalty penalizes large weights much more heavily than L_2

d. The L_1 norm computes the determinant of the covariance matrix

The level sets of the L_1 ball are non-smooth polytopes with vertices on the axes, where elliptical loss contours naturally contact first.

Section 3: Dot Products, Cosine Similarity and Angular Distance

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7 Two query embeddings are $\mathbf{u} = [1, 1, 0]^T$ and $\mathbf{v} = [0, 2, 2]^T$. Compute the cosine similarity $\cos\theta$ between them.

a. 0.500 ✓

b. 0.707

c. 0.250

d. 1.000

$\mathbf{u} \cdot \mathbf{v} = 2$. $\|\mathbf{u}\| = \sqrt{2}$, $\|\mathbf{v}\| = \sqrt{8} = 2\sqrt{2}$. $\cos\theta = \frac{2}{\sqrt{2}(2\sqrt{2})} = \frac{2}{4} = 0.5$.

8 If two unit-normalized embeddings have cosine similarity $\cos\theta = -1.0$, what is their semantic relationship?

a. **Diametrically opposite in direction (180° apart)** ✓

b. Identical in meaning

c. Orthogonal / unrelated

d. Collinear in the exact same direction

$\cos(180^\circ) = -1.0$: the vectors point in exact opposite directions.

9 Compute the cosine distance $D_{\cos}(\mathbf{u}, \mathbf{v}) = 1 - \cos\theta$ for two orthogonal feature vectors.

a. 1.0 ✓

b. 0.0

c. 2.0

d. -1.0

For orthogonal vectors, $\cos\theta = 0$. Thus $D_{\cos} = 1 - 0 = 1.0$.

Section 4: High-Dimensional Geometry and Projections

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10 Find the scalar projection of vector $\mathbf{a} = [4, 2]^T$ onto the direction of unit vector $\mathbf{u} = [0.6, 0.8]^T$.

a. 4.0 ✓

b. 5.0

c. 2.4

d. 3.2

Scalar projection $= \mathbf{a} \cdot \mathbf{u} = 4(0.6) + 2(0.8) = 2.4 + 1.6 = 4.0$.

11 In high-dimensional embedding spaces ($D = 1024$), why are two randomly selected vectors almost certainly nearly orthogonal?

a. **The surface area of a high-dimensional hypersphere concentrates near the equator relative to any pole** ✓

b. All high-dimensional vectors have zero norm

c. The dot product of random Gaussian variables always equals 1

d. The cosine function approaches infinity as dimension increases

By concentration of measure on the sphere $\mathbf{S}^{D-1}$, almost all mass lies in a thin band around the equator, so $\cos\theta \approx 0$ with high probability.

12 Normalize the feature vector $\mathbf{x} = [3, 0, 4]^T$ to have unit Euclidean norm.

a. $[0.6, 0, 0.8]^T$ ✓

b. $[0.75, 0, 1]^T$

c. $[1, 0, 1]^T$

d. $[0.3, 0, 0.4]^T$

$\|\mathbf{x}\| = \sqrt{3^2 + 0^2 + 4^2} = 5$. Unit vector $= \frac{1}{5}[3, 0, 4]^T = [0.6, 0, 0.8]^T$.

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